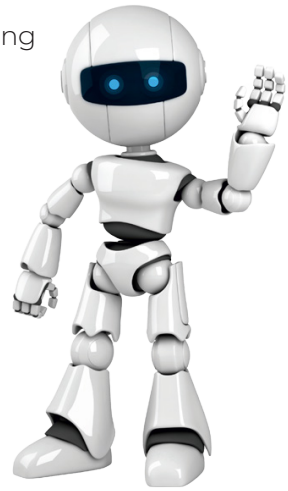
MARKET CORRELATIONS

A guide to understanding these sometimes curious and dynamic market relationships.

On Tuesday April 19, 2022, Bitcoin was trading in the 41000s when abrupt bear volume came in along with an assertive red candle. Bears seemed to have gained confidence out of nowhere, but this was hardly the case. The reality was that on the other side of the market, Netflix had released less than stellar earnings, causing a vicious plummet in stock price. You can probably see where this is going and yes, Bitcoin dropped in value because of poor Netflix earnings. Fundamentally, what does Bitcoin have to do with Netflix? The answer? Absolutely nothing. The reason for the drop is that in the modern world of trading, the market behaves much like a living organism where actions in one area have an effect on other areas, similar to how an injured foot can lead to back pain. The days of completely independent price action across markets and within are largely a thing of the past.

Why Do We See Correlations?

The prevalence of algorithmic trading has been growing as the financial markets mature and at this point estimates have them accounting for 60-75 percent of all trading volume. With robots programmed to instantly react to price action elsewhere and make corresponding trades quicker than a human can react, we end up seeing markets ebb and flow



together even when there is no fundamental tie between instruments. Beyond that, sentiment bleeds through the market leading people to make trading decisions based on price action elsewhere. If the S&P 500 is in a strong uptrend, traders will be more likely to buy something like cannabis stocks even though there are no cannabis holdings in SPY. The same can be said of the opposite, if SPY is seeing a sharp pullback people may sell other assets as to batten up the ship and protect against potential downside. We also see high risk assets correlate together almost as if they are in the same sector even though they may have no fundamental tie. They end up functioning as their own "risk-on" sector of sorts and this can give clues to other areas of the market. If other traders are comfortable buying riskier assets, what's the harm in holding a stalwart name like AAPL for example? Other times, we can see correlations with no notable fundamental or risk level ties, such as when we saw travel-based names trading in lockstep with XLF following the COVID dump.

Chart BTC/USD vs SPY (1m) exhibiting a strong correlation

The Heart Of The Market

Now let's talk about SPY, also known as the heart of the market. The S&P 500 is made up of roughly the top 500 companies traded in the US markets. These companies span across all sectors and their union under SPY gives a solid representation of "average" price movement. When an individual holding moves, it affects SPY based on its weighting within the ETF. Along with affecting SPY, individual stock movement also affect their sector ETFs, such as Apple affecting QQQ - the ETF for the Technology Sector. With that, when trading any stock, you want to keep an eye on its sector ETF, along with the S&P 500. So if a stock's movement affects SPY, its corresponding sector ETF, and people make moves on stocks based on SPY, you can see how this interlocked web of correlations ends up resulting in a market that ebbs and flows as one.

Chart TSLA vs QQQ (5m) exhibiting a bullish correlation

Are Correlations Permanent?

While a high tide raises all ships uniformly, this isn't the case in the financial markets. We see correlations come and go and strengthen and weaken as time goes on and the market's live representation of fear and greed ever changes. This shift and prevalence of bullish and bearish correlations is what creates opportunity for traders. It is a lot easier to trade a name bullish when it is displaying a bullish correlation to its correlational constituents and the same can be said for the opposite side. So far so good, but how do we identify these correlations?

How Do We Identify Correlations?

The answer to this question is all about analyzing peaks and troughs in the price action. If the timing of those peaks and troughs align, great! You've identified two correlated names. Now for the next step, deciphering if you have a bullish or bearish correlation. This information also comes from looking at the peaks and troughs in conjunction with trend analysis. Let's start by looking at the two primary forms of bullish correlation, using TSLA and SPY for a hypothetical example. If SPY is setting a 5min lower high and TSLA sees a 5min higher high, that is one sign of a bullish correlation. The other sign is if SPY sees a 5min lower low while TSLA sees a 5min higher low, in both of these instances you have yourself a bullish correlation on TSLA compared to SPY. Now let's look at a bearish correlation using JPM and the Financial Sector ETF, XLF. If JPM sees a daily lower high while XLF sees a daily higher high, you have yourself a bearish correlation. If JPM sees a daily lower low while XLF sees a daily higher low, you guessed it - bearish correlation. Along with this, we can see inverse correlations such as the Dollar's historical inverse relationship with precious metals. A trader can look for, say, a daily higher low on gold to align with a daily lower high on the Dollar when the inverse relationship is present.

Chart NVDA vs SPY (5m) exhibiting a bearish correlation

How Do We Use Correlations In Our Trading?

Aside from breaches of support and resistance, you can use retracement size in the same fashion as above to help identify bullish and bearish correlations. Say the market has a strong morning before entering afternoon consolidation, if an individual stock retraces 75% of the morning move while another only retraces 20%, the first one has a more bearish correlation and the latter a more bullish. Constantly analyzing these relationships can help us ensure that we are taking trades with the strongest probabilities in our favor. If you are looking for a bullish trade, your odds of avoiding stop out increase when you are playing a name with a bullish correlation to the broad market and vice versa for bear trades.

This brings up another important piece to the puzzle, you do not want to fight the tide of the broad market in most instances. If SPY is seeing a strong pullback, you do not want to be playing a correlated individual name bullish until technical analysis indicates a bounce on SPY is gaining favor. You want the SPY technical analysis and most likely scenario to corroborate what you are looking for on your individual stock. If it does not, you will find yourself swimming against the current.

So we know what correlations are and how they behave, but is this a law of the markets or does it come and go? While the concept of correlations and how they relate is a law of the markets, individual prevalence at any given moment is not. Intraday, one of the biggest causes of correlation shifts is when a key support or resistance holds on an individual name. Bigger picture, the primary thing that will cause correlation shifts are fundamental catalysts which in turn bring in the real game changer - volume. Just as someone would need to have serious power to swim against a strong current, a stock needs volume to move against the current of the market. When high volume is present on an individual name, correlations can fade and more autonomous movement can take place. Conversely, if we see market-wide fear or greed from a catalyst such an FOMC meeting, we will often see correlations strengthen as the market reacts as a whole.

Common pairings to watch and check for correlations...

Individual name	→	Corresponding sector ETF
Individual name	→	SPY
Sector ETF	→	SPY
High risk assets	→	Other high risk assets
Bitcoin	→	QQQ
Bitcoin	→	SPY
Alt Coins	→	Bitcoin
MJ Sector	→	SPY

Utilizing Ratio Charts To Identify Relative Strength/Weakness

What if you want to use quantifiable metrics to identify bullish and bearish correlations? Well, you are in luck! The way to do this is by utilizing what is known as ratio charts. This is simply a chart that plots one ticker's price divided by another ticker's price and the results are invaluable to a trader who knows how to interpret and utilize the information. Essentially, if the ratio chart is moving up, the first name in the ratio has a bullish correlation to the second. If the ratio chart is moving down, it's the opposite - the first name has a bearish correlation to the second. Sounds simple enough but here's the kick, the laws of price movement and technical analysis apply to these ratio charts in the same way they apply to an individual ticker. So you will see 5 minute uptrends, double tops, oversold bounces, all play out on the ratio charts. A trade will have higher odds of success when the ratio chart with its correlated name is telling the same technical story that the individual ticker is telling. Meaning, you want to make bullish trades on names with bullish ratio charts, you want to make bearish trades on names with bearish ratio charts. That is a very complex topic simplified, diving into the charts yourself and piecing together the ever-evolving puzzle of the markets is the fun part.

Further Information

For an in-depth look into market correlations with examples, check out this video...

Video Click to play